# Information Theory-Based Methods (II) Normalized Mutual Information (NMI)

- Mutual information $I(C,G) = -\sum_{i=1}^{r}\sum_{j=1}^{k} p_{ij} \log \frac{p_{ij}}{p_{C_i} p_{G_j}}$
  - Quantify the amount of shared info between the clustering C and the ground-truth partitioning G
  - Measure the dependency between the observed joint probability $p_{ij}$ of C and G, and the expected joint probability $p_{C_i} p_{G_j}$ under the independence assumption
  - When C and G are independent, $p_{ij} = p_{C_i} p_{G_j}$, I(C, G) = 0
  - However, there is no upper bound on the mutual information
- Normalized mutual information $NMI(C,G) = \sqrt{\frac{I(C,G)}{H(C)} \frac{I(C,G)}{H(G)}} = \frac{I(C,G)}{\sqrt{H(C)H(G)}}$
  - Value range of NMI: [0,1]
  - Value close to 1 indicates a good clustering

# Pairwise Comparison-Based Methods: Jaccard Coefficient

- Pairwise comparison: treat each group in the ground truth as a class
- For each pair of objects ($o_i$, $o_j$) in D, if they are assigned to the same cluster/group, the assignment is regarded as positive; otherwise, negative
  - Depending on assignments, we have four possible cases

Note: Total # of pairs of points

$$N = \binom{n}{2}$$

| | $C(o_i) = C(o_j)$ | $C(o_i) \neq C(o_j)$ |
|---|---|---|
| $G(o_i) = G(o_j)$ | true positive (TP) | false negative (FN) |
| $G(o_i) \neq G(o_j)$ | false positive (FP) | true negative (TN) |

- **Jaccard coefficient:** Ignoring the true negatives (thus asymmetric)
  - *Jaccard* = *TP*/(*TP* + *FN* + *FP*) [i.e., denominator ignores *TN*]
    - *Jaccard* = 1 if perfect clustering
- Many other measures are based on the pairwise comparison statistics:
  - Rand statistic
  - Fowlkes-Mallows measure